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Correlation-based structural dropout for convolutional neural networks

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ABSTRACT

Convolutional neural networks (CNNs) easily suffer from the over-fitting problem since they are often over-parameterized in the case of small training datasets. The conventional dropout that drops feature units randomly works well for fully connected networks, but fails to regularize CNNs well due to high spatial correlation of the intermediate features, which allows the dropped information to flow through the network, thus leading to the problem of under-dropping. To better regularize CNNs, some structural dropout methods such as SpatialDropout and DropBlock have been proposed by dropping feature units in continuous regions randomly. However, these methods may suffer from the over-dropping problem by discarding the critical discriminative features, thus limiting the performance of CNNs. To address these issues, we propose a novel structural dropout method, Correlation based Dropout (CorrDrop), to regularize CNNs by dropping feature units based on feature correlation. Unlike the previous dropout methods, our CorrDrop can focus on the discriminative information and drops features in a spatial-wise or channel-wise manner. Extensive experiments on different datasets, network architectures, and various tasks (e.g., image classification and object localization) demonstrate the superiority of our method over other methods.

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1. Introduction

Convolutional neural networks (CNNs) have been widely used in machine learning community and computer vision tasks, including image recognition [1] and object detection [2–4]. In recent years, many advanced CNNs like ResNet [5], InceptionNet [6] and DenseNet [7] are designed to improve the performance of the conventional CNNs. Deeper and wider deep learning models [8] are proposed to achieve state-of-the-art performance in various computer vision tasks. However, these models have millions of parameters and thus easily suffer from the problem of over-fitting, especially in the case of limited training data. Therefore, developing regularization methods to relieve the over-fitting of CNNs is essential.

There are numerous regularization methods proposed early, such as weight decay [9], early stopping [10], data augmentation [11,12], batch normalization [13,14] and dropout [15]. These methods have been adopted as the conventional toolkits to regularize deep neural networks. Among them, the traditional dropout works well for fully connected networks (FCs). However, such dropout can not effectively regularize CNNs by dropping single feature unit randomly in the feature map, since the spatially correlated features still allow dropped information to flow through the network leading to the under-dropping problem.

In order to make dropout more effective for CNNs, some structural dropout methods, including SpatialDropout [16], Cutout [17] and DropBlock [18], have recently been proposed to improve the regularization ability for CNNs. These methods attempt to regularize CNNs by randomly dropping entire channels or square of regions in the input/feature space. However, they suffer from the over-dropping problem as the feature units are dropped in continuous regions in a random manner without considering the semantic information in the images. Such random manner of dropping feature units may discard the whole of discriminative regions in the input/feature maps and limit the learning capability of the models. As shown in Fig. 1, the traditional dropout [15] drops the single unit in the feature maps while the structural DropBlock [18] drops a square of feature units in the feature maps directly and may zero out the informative semantic regions.

Inspired by the observation that discriminative region of an object would have higher feature correlations (see the last row in

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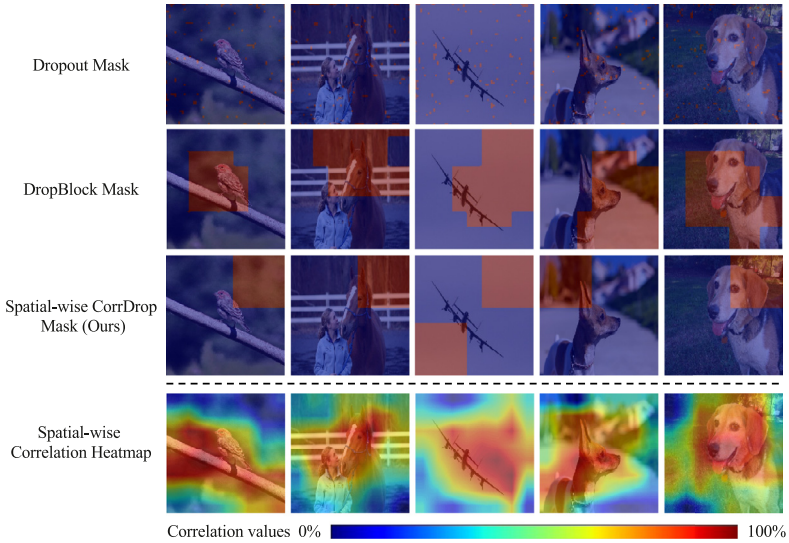

Fig. 1. Examples of Dropout [15], DropBlock [18] and our Spatial-wise CorrDrop masks (top three rows). The red regions denote the regions to be masked. The last row shows the corresponding correlation heatmap of CorrDrop. The feature units focusing on the main objects are more correlated with each other. Compared with Dropout and DropBlock, CorrDrop takes the discriminative information into consideration and drops feature units adaptively to alleviate the under- and over-dropping problems. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Fig. 1), we propose Correlation-based Dropout (CorrDrop), a novel and effective structural dropout method for CNNs, which drops feature units considering the feature correlation in spatial/channel dimensions. Unlike the previous structural dropout methods (e.g., DropBlock) that drop feature units randomly, our CorrDrop drops the feature units adaptively based on the discriminative information. Specifically, we first compute the feature correlation map to indicate the most discriminative regions, and then adaptively mask out those less discriminative regions, i.e., regions with small feature correlation values. Since feature correlation can be further divided into spatial-wise and channel-wise feature correlation based on the way of correlation computation, we propose two variants of CorrDrop: Spatial-wise CorrDrop and Channel-wise CorrDrop, which adaptively drops features in spatial dimension and channel dimension, respectively. As shown in Fig. 1, compared with the conventional dropout and DropBlock that encounter the under-/over-dropping problems, our CorrDrop produces adaptive masks by discarding part of less correlated regions, thus making the network learn more compact representations. Extensive experiments on image classification demonstrate that our CorrDrop consistently outperforms dropout [15] and DropBlock [18] under different architecture of CNNs on public datasets. Furthermore, we demonstrate that our CorrDrop also regularizes CNN models well in other computer vision tasks such as object localization.

A preliminary version of this work has been presented as a conference version [19]. In this extension version, we incorporate additional content including the channel-wise CorrDrop, more ablation study, experiments on the state-of-the-art CNNs and additional vison tasks. The main contributions could be summarized as follows:

- We proposed Correlation based structural dropout (CorrDrop) that drops the less correlated features in the feature maps which alleviates the under-/over-dropping problems of previous dropout variants that drop features in a random manner.
- With regard to the spatial-wise and channel-wise features in the feature map, the corresponding Spatial-wise CorrDrop (SCD) and Channel-wise CorrDrop (CCD) are proposed. Experimental results show that they work in complementary that SCD performs well on simple datasets (e.g. CIFAR-10 and SVHN) while CCD is superior on complex datasets (e.g. CIFAR-100 and Tiny-ImageNet).
- Extensive experiments on various datasets, architectures and vision tasks demonstrate that our methods can get consistent improvement.

The remainder of this paper is organized as follows. In Section 2, we briefly review the related works on regularization methods and attention mechanism in deep learning. In Section 3, we introduce the proposed CorrDrop in detail. Then, experimental results are provided in Section 4. Finally, we draw the conclusions in Section 5.

2. Related works

Regularization in Deep Learning. Deep neural networks can be easily over-fitted in practice due to the huge amount of parameters, thus hindering the generalization of models. To address such problem, many regularization methods [9,13,20] have been proposed in the past few years to improve the performance of deep learning models. Among them, weight decay [9] adds an additional term in the loss function as a function of the norm of parameters in order to limit the capacity of the model. Data augmentation [20,21] randomly applies the operations like rotation, reflection and flipping to the input images to supplement the data insufficiency problem in deep learning for visual tasks. Deep feature augmentation [22] has recently been proposed to enhance the robustness of features for occluded image classification. Early stopping [10] is another easy-implemented method to avoid overfitting through stopping the training when the test loss begins to increase. Batch normalization (BN) [13] normalizes the input features in each layer to zero means and unit variances. BN can reduce the internal covariate shift and avoid gradient vanish so as to help the model better converge. Meanwhile, the noise induced in the computation of the means and variances in each mini-batch can also regularize the network. Beyond so many regularization methods, dropout [15] is one of the most widely used regularization method to improve the performance of deep neural networks. The basic idea of dropout is to randomly set hidden units to zero with a certain probability during training. In this way, it can reduce the dependence of features and improve the performance by

ensembling different architectures in an implicit way. Since then, many regularization methods borrow the idea from dropout by injecting noise into neural networks to alleviate the over-fitting such as DropConnect [23], StochasticDepth [24], DropPath [25] and shake-shake regularization [26]. The main drawback of the traditional dropout lies in that it fails to work well for CNNs due to the under-dropping problem [18]. Recently, several structural dropout methods such as SpatialDropout [16], Cutout [17] and DropBlock [18] have been proposed to enhance the regularization effect on CNNs. Among them, SpatialDropout randomly drops entire channels in feature maps, while Cutout randomly masks out square regions in the input. Inspired by Cutout, recently proposed DropBlock generalizes Cutout in the feature space by masking out square feature regions randomly. However, these regularization methods can not always work well and may have the risk of the over-dropping problem, as they may discard the most discriminative regions by dropping square regions of input/feature randomly. Instead, our proposed CorrDrop alleviates these problems by dropping feature units adaptively based on feature correlation that reflects the discriminative information of features. Similarly, the recent proposed Guided Dropout [27] also considers dropping nodes by measuring the strength of each node instead of random dropping. Compared to Guided Dropout, our method measure the importance of feature units by feature correlation and also combine with the structural dropout.

Attention Mechanism. Attention mechanism has been developed inspired by the phenomenon that humans tend to focus on the discriminative parts of the images. Various attention methodologies have been widely applied in different fields such as machine translation [28], image classification [29], transfer learning [30], and weakly supervised object localization [31]. In general, attention mechanism can be divided into two types: spatial-wise and channel-wise attention. Among them, non-local neural networks [32] is a typical method of spatial-wise attention method that computes the response at a position as a weighted sum of the features at all positions in the input feature maps. In contrast, squeeze-and-excitation network (SE-Net) [33] is a channel-wise attention method and adaptively rescales channel-wise feature responses by explicitly modelling inter-dependencies between channels. Recent proposed method [34] exploits spatial and channel inter-dependencies to obtain meaningful features for image super-resolution.Attention mechanism focuses on improving the representational ability of models by explicitly/implicitly utilizing the feature correlation. Unlike the attention-based methods that improve the representational ability, our CorrDrop focuses on regularizing models effectively based on feature correlation. Besides, our CorrDrop does not add any additional trainable parameters and can combine with attention mechanism to further improve the performance.

3. Methodology

Most existing dropout-based methods are limited in regularizing CNNs due to the problems of the under-/over-dropping. By exploiting the correlation of features, we propose an effective structural dropout: correlation-based dropout (CorrDrop), which drops the feature units adaptively based on the discriminative information and can effectively regularize CNNs. Considering the spatial-wise and channel-wise feature correlation of CNNs, we can further derive two variants of CorrDrop, that is Spatial-wise CorrDrop and Channel-wise CorrDrop. The pipeline of these two variants are shown in Fig. 2 and 3. In the following sections, we firstly describe the calculation of feature correlation based on feature orthogonality. Then we sample the mask according to the correlation map. Finally, we illustrate the strategies of Spatial-wise CorrDrop and Channel-wise CorrDrop.

3.1. Feature correlation calculation

Unlike the previous methods that drop feature units randomly, we attempt to drop feature units adaptively based on feature correlation, which reflects discriminative information. The recent study [35,36] has shown that discriminative regions of an object would have higher feature correlations. Such observations let us make the basic assumption that it is more effective in regularizing CNNs by dropping those low-correlated features. To represent feature correlation, we use the metric with feature orthogonality as in the previous work [37]. Given the feature matrix $A = [a_1, \ldots, a_N]^T \in \mathbb{R}^{N\times K}$, where N is the number of feature units, K is feature dimension. The correlation calculation can be described as below,

$$\widehat{A} = \frac{A}{\|A\|}, \tag{1}$$

$$P = |\widehat{A} \times \widehat{A}^T - I|, \tag{2}$$

$$F_i = \frac{\sum_N P_i}{N}, \tag{3}$$

where $|\cdot|$ denotes absolute value operation, I is an identity matrix with size $N \times N$. We firstly normalize each row of A and compute the correlation scores based on the feature orthogonality. P is a matrix of size $N \times N$ and P_i denotes i-th row of P. Off-diagonal elements of a row of P for a single unit denote projection of all the other feature units. The mean of each row denotes the correlation score of each unit. The higher value of F_i indicates that this unit is highly correlated with others.

3.2. Correlation based dropout mask sampling

To drop units adaptively based on the feature correlation, we assign the dropout probability to each unit according to the values in F. Generally, the higher value of F_i is, the smaller dropout probability we have. The dropout probability of i-th feature unit in A can be formulated as,

$$\gamma_i = 1 - \frac{e^{F_i}}{\sum_{j=1}^{N} e^{F_j}}, \tag{4}$$

where i and j denote the index of feature unit in F. To ensure the dropout probability $\gamma_i \in (0, 1)$, we normalize the correlation score of each unit.

The dropout mask $M \in \mathbb{R}^N$ is sampled from Bernoulli distribution based on dropout probability γ_i,

$$M_i = Bernoulli(1 - \gamma_i). \tag{5}$$

Empirically, similar to other dropout variants, a hyper-parameter p is introduced to ensure that our CorrDrop would not drop too many feature units. With the correlation based dropout mask M, we adjust the keep probability and generate another mask $B \in \mathbb{R}^N$. The units are dropped when the corresponding values in both two masks are 0 and the final dropout mask $S \in \mathbb{R}^N$ is obtained. The final dropout mask of CorrDrop can be formulated as follows,

$$\widehat{p} = \frac{p \times numel(M)}{numel(M) - sum(M)}, \tag{6}$$

$$B_i = Bernoulli(1 - \widehat{p}), \tag{7}$$

$$S_i = \begin{cases} 0 & M_i = 0 \ and \ B_i = 0, \\ 1 & otherwise \end{cases} \tag{8}$$

where $numel(M)$ counts the number of units in M, $sum(M)$ counts the number of units where the value is 1.

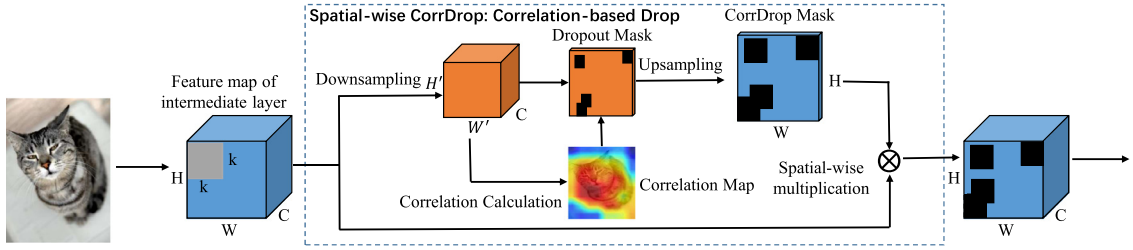

Fig. 2. The pipeline of Spatial-wise CorrDrop. 1) Downsample the feature maps from previous layers by spatial-wise local average pooling with kernel size and stride of k for local features gathering and dimension reduction. 2) A correlation map is calculated based on the feature orthogonality, and the dropout mask is sampled from a Bernoulli distribution with adaptive dropout probability. 3) The CorrDrop mask is generated by nearest neighbour upsampling. 4) The regularized feature is obtained by performing element-wise multiplication of the CorrDrop mask and the original feature map.

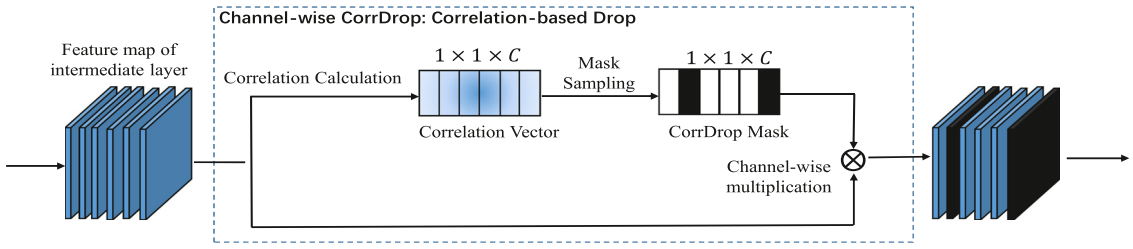

Fig. 3. The pipeline of Channel-wise CorrDrop. 1) Calculate the correlation vector based on the correlation calculation. 2) Sample the CorrDrop mask according to the correlation vector, that is the less correlated channels are more likely to be dropped. 3) Channel-wise multiplication is performed to multiply the CorrDrop mask to the original feature map.

3.3. Spatial-wise CorrDrop

In spatial dimension, we suppose that highly correlated units construct the discriminative parts in the feature maps, which should be kept with higher probability. Given the feature maps of the intermediate l-th layer as $V^{(l)} = [v_1^{(l)}, \dots, v_N^{(l)}]^T \in \mathbb{R}^{N\times C}$, where $N = H \times W$ is the number of units in a feature map, C is the number of channels, H and W are the height and width of the feature map respectively. Each row $v_i^{(l)} \in \mathbb{R}^{\mathbb{C}}$ represents the feature vector of a unit. As the features are locally correlated in CNNs, it is less effective to drop a single unit in the feature map [18]. Following the previous work DropBlock [18] which drops continuous regions in the feature map, we further consider the correlation of each local area and drop blocks of units. To obtain a structural mask, we firstly gather the local information in the feature map by local average pooling and meanwhile reduce the dimension of the feature map for speeding up the correlation calculation. When setting the block size to k, we perform local average pooling on the feature map with kernel size of k and stride of k. Specifically, we scan each block with size of $k \times k$ from left to right, top to bottom in each feature map and compute the mean of activation values in each block, which can be described as

$$v_{i',j'}^{(l)'} = \frac{\sum_{k_1=-\frac{k}{2}}^{\frac{k}{2}} \sum_{k_2=-\frac{k}{2}}^{\frac{k}{2}} v_{i+k_1,j+k_2}^{(l)}}{k^2}. \tag{9}$$

The resulted feature map is $V^{(l)'} \in \mathbb{R}^{N'\times C}$, where $N' = H' \times W'$, $H' = ceil(\frac{H}{k})$, $W' = ceil(\frac{W}{k})$. The dropout probability p is adjusted accordingly as,

$$p' = \frac{p}{k^2} \frac{H \times W}{H' \times W'}. \tag{10}$$

With the downsampled feature map $V^{(l)'}$, we sample the corrdrop mask as,

$$\begin{aligned} F_s^{(l)} &= \Phi(V^{(l)'}), \\ S_s^{(l)'} &= \Psi(F_s^{(l)}, p'), \end{aligned} \tag{11}$$

where $\Phi(\cdot)$ is the function of feature correlation calculation as in Equ. (1)-(3) and $\Psi(\cdot)$ denotes the dropout mask sampling operation as in Equ. (4)-(8). To generate the structural mask, we take advance of nearest neighbour upsampling to upsample the corrdrop mask $S_s^{(l)'} \in \mathbb{R}^{H'\times W'}$ to $S_s^{(l)} \in \mathbb{R}^{H\times W}$. Each zero entry in $S_s^{(l)}$ will be expanded to $k \times k$ block. Thus, square regions of units will be dropped. Finally the spatial-wise corrdrop mask is multiplied to each channel of the original feature maps $V^{(l)}$ and masks out part of feature regions, which denoted as,

$$\widetilde{V}^{(l)} = S_s^{(l)} \odot V^{(l)}, \tag{12}$$

where $\odot$ represents the point-wise multiplication operation. The pipeline is shown in Fig. 2. In this manner, we calculate the feature correlation based on the local information and drop square of regions with small average correlation.

3.4. Channel-wise CorrDrop

In addition to the spatial-wise features, it is worth noting that each CNN filter can detect different patterns of the input data, that is channel-wise features correspond to different semantic patterns. The work in [38] indicates that more semantic feature channels are more class-specific with some of redundant and less activated channels. Therefore, we attempt to drop those uncorrelated feature channels based on channel-wise feature correlation and improve the generalization ability, leading to our Channel-wise CorrDrop. Similar to Spatial-wise CorrDrop, given the feature maps of the intermediate l-th layer as $V^{(l)} = [v_1^{(l)}, \dots, v_N^{(l)}]^T \in \mathbb{R}^{N\times C}$, where $N = H \times W$ is the number of units in a feature map, C is the number of channels, H and W are the height and width of the feature map respectively. We first reshape the l-th layer feature map $V^{(l)}$ to $U^{(l)} = [u_1^{(l)}, \dots, u_C^{(l)}]^T \in \mathbb{R}^{C\times N}$. Similarly, the channel-wise dropout mask is computed as,

$$\begin{aligned} F_c^{(l)} &= \Phi(U^{(l)}), \\ S_c^{(l)} &= \Psi(F_c^{(l)}, p), \end{aligned} \tag{13}$$

where p is dropout probability, $F_c^{(l)} \in \mathbb{R}^C$ is correlation map, $S_c^{(l)} \in \mathbb{R}^C$ is corrdrop mask. The channel-wise corrdrop is performed as,

$$\widetilde{U}^{(l)} = S_c^{(l)} \odot U^{(l)}, \tag{14}$$

where $\odot$ refers to channel-wise multiplication, and the j-th channel of $U^{(l)}$ will be set to zero if the j-th element in $S_c^{(l)}$ is zero.

4. Experiments

To evaluate the effectiveness of our CorrDrop, we compare it with other state-of-the-art dropout-based methods, including the conventional Dropout [15], SpatialDropout [16], DropBlock [18] and Cutout [17]. Firstly, extensive experiments are conducted to analyze the effects of hyper-parameters for CorrDrop. Then, various experiments are conducted on image classification task under benchmark datasets (e.g., CIFAR-10 [39], CIFAR-100 [39], SVHN [40] and Tiny-ImageNet [41]) and network architectures (e.g., ResNet [5], VGG [42] and DenseNet [7]). Besides the regularization effects on CNNs, our CorrDrop can combine with other attention strategies to further improve the performance. To have an intuitive understanding for CorrDrop, we analyze the feature activation in terms of quantitative analysis and activation visualization. Finally, we demonstrate the potential applications of CorrDrop on other vision tasks such as weakly supervised object localization.

4.1. Experimental settings

For image classification, we perform our experiments on CIFAR-10, CIFAR-100 [39], SVHN [40] and Tiny-ImageNet [41]. CIFAR dataset consists of 60,000 color images of size 32×32 with 10 classes for CIFAR-10 and 100 classes for CIFAR-100. The Street View House Numbers (SVHN) dataset consists of house numbers with 10 classes of size 32×32. Tiny-ImageNet has 200 classes in total, and the size of each image is 64×64, in which each class has 500 training images, 50 validation images and 50 test images.

We normalize the input data to have a zero mean and unit variance. During training, we adopt the standard data augmentation schemes like flipping and random cropping. For fair comparison, we report the best results of the compared methods. The default setting of batch size is set to 128, weight decay is 1e-4. and stochastic gradient descent (SGD) with momentum of 0.9 is used to optimize the model parameters. Unless otherwise specified, experiments on CIFAR datasets are tested on ResNet20 [5] using the official PyTorch implementation. Total training epoch is 200, and the initial learning rate is 0.1 which are reduced by a factor of 10 at 80th, 120th, 160th epoch. For Tiny-ImageNet, we conduct the experiments on ResNet18, total training epoch is 90, the initial learning rate is 0.1 and decayed by a factor of 10 at 30th and 60th epoch. For SVHN, we train ResNet20 for 160 epochs, initial learning rate is 0.01 and decayed by factor of 10 at 80th and 120th epoch. For dropout probability, as shown in the work [18], we gradually increase the value of dropout probability p with linear scheduler. For simplicity, we denote our Spatial-wise CorrDrop and Channel-wise CorrDrop as **SCD** and **CCD**, respectively.

4.2. Hyper-parameter choice of CorrDrop

In this section, we test the effects of hyper-parameters in our methods, which mainly includes dropout probability p, block size k and embedded stage.

Choice of dropout probability p**.** In Fig. 4(a), we display the test accuracy on CIFAR-10 with Dropout, DropBlock, SCD and CCD with varying dropout probability. From Fig. 4(a) we can observe that SCD and CCD obtain the best performance around $p = 0.2$. Compared with other methods, SCD/CCD achieve better performance in most of different settings, which verifies the superiority of our SCD/CCD over other regularization methods. Note that the performance of CCD degrades quickly when the dropout probability p is larger than 0.3. This is because large p indicates that many feature channels would be dropped, thus leading to the loss of too much semantic information.

Table 1
Effect of integrating SCD/CCD blocks in ResNet20 at different stages on CIFAR-10 (Top-1%).

Stages	SCD	CCD
Baseline	91.61	91.61
Stage1&2 (Shallow layer)	91.76	91.90
Stage2&3 (Deep layer)	**92.45**	**92.42**
ALL Stages	92.39	92.18

Table 2
Accuracy on CIFAR-10 (%) when integrating SCD/CCD blocks into different location of the residual module.

Stages	Spatial-wise CorrDrop	Channel-wise CorrDrop
Baseline	91.61	91.61
Residual-PRE	92.41	91.99
Residual-POST	**92.45**	**92.42**
Identity-POST	91.98	91.02

Choice of block size k**.** For SCD, block size k also plays a critical role. To test the effects of k, we embed SCD and DropBlock blocks into the last two convolutional groups (stage2 and stage3) of ResNet20 with different choices of k. The results are shown in Fig. 4(b). Notice that the baseline model is trained without SCD or DropBlock. From Fig. 4(b) we see that SCD outperforms the baseline model under varying k. Furthermore, compared to DropBlock, SCD can obtain better results in most cases. Empirically, SCD yields the best performance around $k = 5$. Thus, in our default setting, we set the block size k to 5.

Choice of embedded stage. To test the choice of embedded stages of SCD/CCD, we test experiments with SCD/CCD embedded into different stages on ResNet20. Specifically, ResNet20 in our experiments has three groups of convolutional layers (stage1, stage2, stage3). Each stage consists of six 3×3 convolutional layers after each of which the size of feature map is halved with a convolution with a stride of 2. We choose two stages at a time. Specifically, we choose stage1&2, stage2&3 to represent the shallow and deep layers. The results are reported in Table 1. Experimental results show that the embedded stages of SCD/CCD influence the performance, and the embedded stages in deeper layers obtains significantly better results than that in shallow layers. This is because the features in deeper layers contain more semantic information of object. Additionally, we also embeded SCD/CCD into all stages, but did not bring extra performance improvement. Therefore, in our default setting, we embed SCD/CCD blocks into the last two stages of the networks.

Combination with residual block. It is known that residual block builds the basic blocks for many state-of-the-art CNN models. Here, we conduct experiments to analyze how our SCD/CCD blocks are integrated into the residual block. We conduct experiments on CIFAR-10 with ResNet20. As depicted in Fig. 5, we consider three variants of how SCD/CCD is integrated into residual block. The conventional residual module consists of a residual branch and identity mapping branch. Specifically, we obtain three variants: (1) Residual-PRE: SCD/CCD blocks are inserted before the residual block; (2) Residual-POST: SCD/CCD blocks are inserted after the residual block; (3) Identity-POST: SCD/CCD blocks are inserted after the sum of identity mapping branch and residual branch. The experimental results are reported in Table 2. We observe that Residual-PRE and Residual-POST perform similarly for SCD, while Residual-POST performs better for CCD. In addition, Identity-POST leads to performance degradation for CCD. The experimental results show that CCD block is more sensitive to the location where it is inserted. This is because the discard of the entire channel would lead to the significant loss of more informa-

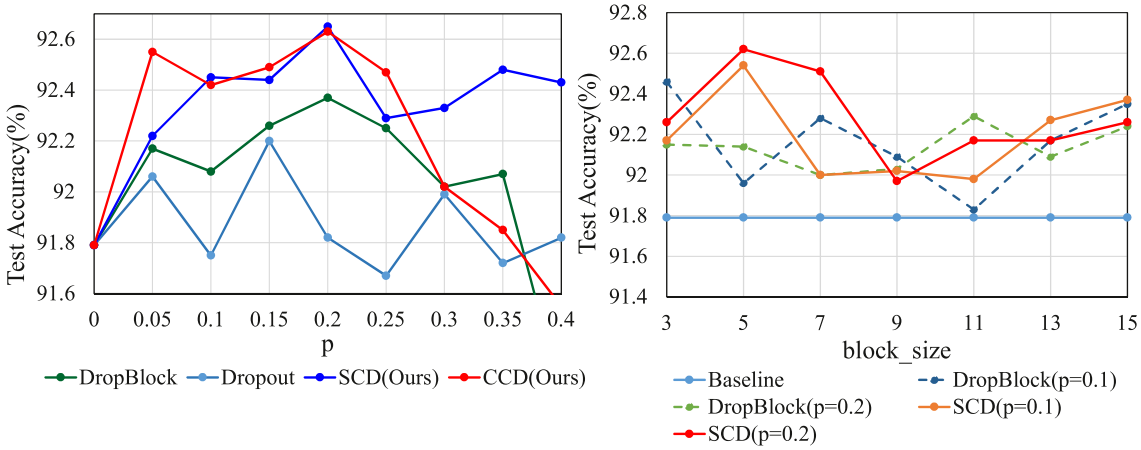

Fig. 4. Subfigure (a) is the test accuracy on CIFAR-10 with different choices of dropout probability. Subfigure (b) is the test accuracy on CIFAR-10 with different choices of block size.

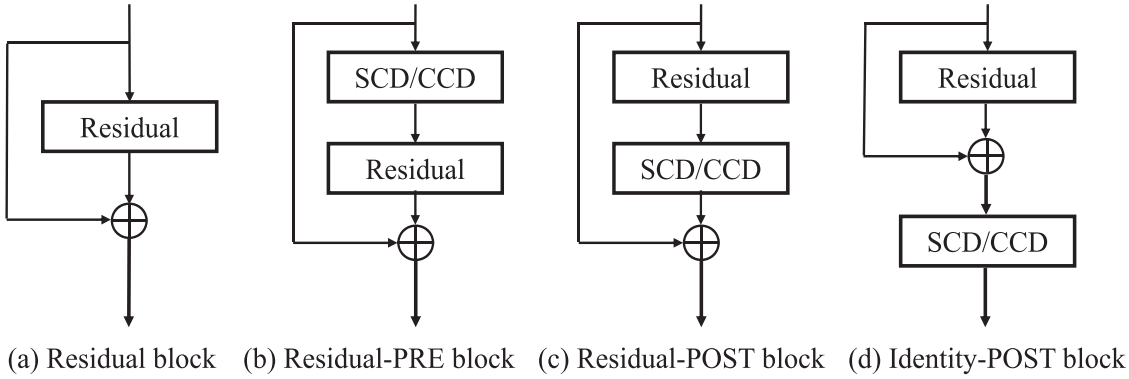

Fig. 5. Combination with residual block.

Table 3
Top-1 (%) classification accuracy on CIFAR-10 dataset with different combination of BN with SCD/CCD.

Combination Order	CIFAR-10	Combination Order	CIFAR-10
Conv-SCD-BN-ReLU	92.38	Conv-CCD-BN-ReLU	92.05
Conv-BN-SCD-ReLU	**92.66**	Conv-BN-CCD-ReLU	**92.50**

tion. Residual-POST performs consistently well for both SCD and CCD blocks.

Moreover, each residual block consists of a convolutional layer, batch normalization layer (BN) and activation layer (e.g. ReLU). As shown in Table 3, we also investigate the effects of SCD/CCD integrated with BN layer. Experimental results show that SCD/CCD works better to embed SCD/CCD after BN layer. When SCD/CCD is put before BN, the dropped features may lead to inaccurate approximation of feature means and variances.

4.3. Comparison with other regularization methods

We compare the regularization effects of our SCD/CCD with other state-of-the-art regularization methods, including the conventional Dropout [15], SpatialDropout [16], DropBlock [18] and Cutout [17]. The experimental settings are described in Section 4.1. We conduct each experiment three times and report the mean accuracy and standard variance in Table 4, from which we can see that both our SCD and CCD obtain very similar results and consistently obtain better performance than other regularization methods on different datasets. For example, compared with baseline, the performance gain of DropBlock on CIFAR-100 is only up to 0.37%, while the performance gain of our SCD and CCD is up to 1.27% and 1.92%, respectively. This is mainly because DropBlock undergoes the risk of over-dropping by dropping square feature regions randomly, while our methods alleviate this problem by adaptively dropping feature based on feature correlation. Moreover, Cutout also works well to regularize CNNs and is usually adopted as a data augmentation technique in practice. It can be cast as a special regularization method by dropping units in the input/image space while our methods regularize CNNs in the feature space. Cutout works well on CIFAR-10, CIFAR-100 and SVHN, while failing to work well on Tiny-ImageNet. Besides, our methods can combine with Cutout to further improve the performance. For example, when SCD/CCD is combined with Cutout, the test accuracy on different datasets can be further improved. From Table 4 we can also observe that SCD performs well on simple datasets such as CIFAR-10 and SVHN, while CCD shows its superiority on harder datasets such as CIFAR-100 and Tiny-ImageNet, which indicates the complementary roles of SCD and CCD in practice. Empirically, CCD serves as a stronger regularizaiton methods. As the wide range of image types and resemblance among classes, classification on CIFAR-100 and TinyImageNet are more challenging tasks. For example, each class in CIFAR-10 contains around 5000 images on average, while in CIFAR-100 and Tiny ImageNet, the quantity plunges to only 500. Overfitting is more severe given the limited training data in CIFAR-100 and Tiny-ImageNet thus powerful regularization methods such as CCD performs better. Finally, we also plot the test accuracy curves on CIFAR-10 and CIFAR-100 with different regularization methods during training in Fig. 6, from which we see that both our SCD and CCD achieve the best results, which further demonstrates the effectiveness of our methods.

Table 4
Top-1 (%) classification accuracy on CIFAR-10, CIFAR-100, Tiny-ImageNet and SVHN with different regularization methods. The value of p is the best parameter decided by grid search. Each experiment is conducted three times and we report the mean accuracy and standard variance.

Methods	CIFAR-10	CIFAR-100	Tiny-ImageNet	SVHN
Baseline (without regularization)	91.61 ± 0.20	67.65 ± 0.42	62.33 ± 0.34	95.37 ± 0.04
Dropout ($p = 0.15$)	91.96 ± 0.23	67.76 ± 0.33	62.78 ± 0.30	96.13 ± 0.10
SpatialDropout ($p = 0.1$)	91.94 ± 0.06	67.93 ± 0.21	63.72 ± 0.38	95.91 ± 0.03
DropBlock ($p = 0.2$)	92.24 ± 0.13	68.02 ± 0.09	62.86 ±0.24	96.04 ± 0.13
SCD (Ours) ($p = 0.2$)	**92.66 ±0.11**	68.92 ± 0.55	63.48 ± 0.42	**96.49 ±0.10**
CCD (Ours) ($p = 0.2$)	92.50 ± 0.12	**69.57 ±0.25**	**64.35 ±0.26**	96.14 ± 0.05
Cutout	92.38 ± 0.18	68.99 ± 0.12	62.16 ± 0.27	96.67 ± 0.10
Cutout + SCD(Ours)($p = 0.03$)	**92.92 ±0.05**	**69.11 ±0.52**	63.15 ± 0.32	**97.05 ±0.08**
Cutout + CCD(Ours)($p = .1$)	92.65 ± 0.24	**69.11 ±0.13**	**63.58 ±0.37**	96.90 ± 0.04

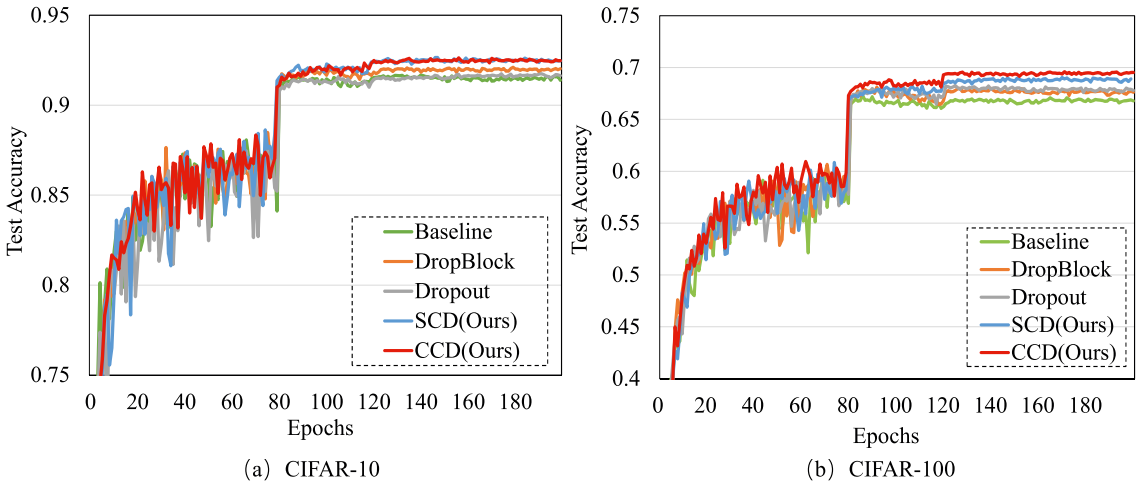

Fig. 6. Test accuracy curves on (a) CIFAR-10 and (b) CIFAR-100 with different regularization methods during training.

Table 5
Top-1 (%) classification accuracy on CIFAR-100 and Tiny-ImageNet datasets compared with Guided Dropout.

Methods	CIFAR-100	Tiny-ImageNet
Baseline without dropout	77.24	63.99
Guided Dropout	77.52	64.33
SCD	77.84	64.61
CCD	**78.14**	**65.71**

Comparison with Guided Dropout. In Table 5, we also provide the experimental results compared to Guided Dropout [27]. We follow the experimental settings as Guided Dropout. We adopt the ResNet18 as backbone and train it for 200 epochs on CIFAR-100 and Tiny-ImageNet. It can be observed that our SCD/CCD outperform Guided Dropout while CCD gives the best performance. Though Guided Dropout also considers the importance of the feature units, it does not consider structurally dropping the features in CNNs.

4.4. Comparison with attention-based methods

In this section, we compare SCD and CCD with non-local network [32] and SE network [33], which are two representative spatial-wise and channel-wise attention methods. We conduct experiments of image classification on CIFAR-10 with ResNet20. SCD and CCD are inserted into the last two convolutional groups (stage2 and stage3). Experimental results are summarized in Table 6, from which we see that our SCD and CCD outperform these two attention-based methods. Furthermore, the gains induced by SCD/CCD and attention-based blocks are complementary, which indicates that they can be combined effectively to further boost the performance. Specifically, the combination of SCD and spatial-wise attention (denoted as SCD+Non-lcoal) obtains the best performance of 92.94%.

Table 6
Comparison with attention based methods. Top-1(%) accuracy on CIFAR-10, computational complexity FLOPs (M) and parameter overhead Params (M) are reported.

Methods	Accuracy	FLOPs(M)	Params(M)
Baseline	91.79	41.22	0.2725
Non-local	91.82	44.53	0.3045
SE	92.04	41.27	0.2802
SCD (Ours)	**92.45**	41.34	0.2725
CCD (Ours)	92.42	42.97	0.2725
SCD + Non-local	**92.94**	44.65	0.3045
SCD + SE	92.89	41.39	0.2802
CCD + Non-local	92.27	46.29	0.3045
CCD + SE	92.61	43.03	0.2802

Analysis of computation and parameters overhead. The computation overhead of our methods mainly come from the calculation of correlation map. Due to the use of a local average pooling operation, the size of the feature map is reduced. The number of FLOPs in the training process only increases by 0.29% for SCD and 4.25% for CCD, compared to baseline. In the test process, SCD is very similar to dropout, and thus the time complexity of SCD would not increase in the test process. In addition, the calculation of correlation map based on the feature orthogonality does not add any additional trainable parameters.

4.5. Regularization on different architectures of CNNs

To demonstrate the generalization of our methods, we compare our methods with other regularization methods on different architectures such as VGG16 [42], ResNet110 [5], DenseNet [7] and Wide

Table 7
Classification accuracy (Top-1%) of different architectures regularized by DropBlock, Cutout and our methods on CIFAR-10.

Methods	VGG16_BN	ResNet110	DenseNet -BC-100-12	Wide-ResNet -28-10
Baseline	93.51	93.25	95.32	95.98
DropBlock	94.05	93.12	95.49	96.26
CCD (Ours)	93.75	93.77	95.30	96.26
SCD (Ours)	94.18	94.27	95.40	95.46
Cutout	94.68	94.53	95.98	96.78
Cutout + SCD (Ours)	**94.74**	**94.76**	**96.33**	**96.95**

Residual Network (WRN). Specifically, VGG16 incorporates batch normalization layers [13] (denoted as VGG16_BN), and DenseNet-BC has 100 layers in total and sets grow rate to 12 (denoted as DenseNet-BC-100-12). WRN has 28 layers and sets widening factor to 10 (denoted as WRN-28-10). The training strategies of VGG16 and ResNet110 are identical to ResNet20 as illustrated in Section 4.1. For DenseNet, the total training epoch is 300 and the initial learning rate is set to 0.1, which is reduced by a factor of 10 at the 150th and 225th epoch. For WRN-28-10, weight decay is set to 5e-4 and the initial learning rate is set to 0.1, which is reduced by a factor of 5 at the 60th, 120th, 160th epoch during training. For fair comparison, the dropout probability is $p = 0.2$, block size for SCD and DropBlock is $k = 5$ and the embedded location of different regulariztion methods are identical for different architectures of CNNs. The classification accuracy results of different models on CIFAR-10 are reported in Table 7. Compared with DropBlock, our SCD consistently achieves better performance under different models, which verifies the effectiveness of our SCD. Note that our CCD obtains very similar results with DropBlock, and sometimes achieves a little worse results than DropBlock. This is because the regularization ability of CCD is too strong for CIFAR-10. However, CCD performs better for those harder datasets such as CIFAR100 and Tiny-ImageNet (see Table 4). Note that the performance of our method can be further improved by 0.06%-0.35% when combining with Cutout, which usually acts as a data augmentation technique.

4.6. Transfer learning results

In recent years, various promising CNN architectures are proposed, and achieve state-of-the-art performance on ImageNet [43]. In this section, we apply our methods on those promising CNN models including BiT [44], FixResNet [45] and ResNeSt50 [46]. We follow the transfer learning paradigm and finetune the ImageNet pre-trained models on CIFAR-10 and CIFAR-100 datasets. To match the input resolution of the models, we resize the image of CIFAR dataset to 128×128 for BiT and ResNeSt, and resize the image of CIFAR dataset to 224×224 for FixResNet. The setting of hyper-parameter follows the BiT-HyperRule in the original paper for BiT-M model. The initial learning rate are 1e-3 and 3e-3 for FixResNet50 and ResNeSt50, respectively. The total training epoch is 80, and the batch size is 64. The experimental results are shown in Table 8, from which we can see that the performance of advanced models can be further improved by integrating our SCD/CCD. For example, CCD outperforms the baseline of ResNeSt50 by 1.74% on CIFAR-100. These experimental results demonstrate that our methods can work as a general regularization technique and can be easy to plug in to existing state-of-the-art CNN models.

4.7. Feature activation analysis

In order to better understand the effects of our CorrDrop, we quantitatively compare the magnitude of the feature activation of the model trained with and without CorrDrop. Then we visualize the activation map with class activation mapping (CAM) [35].

Table 8
Transfer learning results on CIFAR-10/100 datasets (Top1-%) with promising CNN models pretrained on ImageNet.

Models	Regularization	CIFAR-10	CIFAR-100
BiT-M-R50x1	Baseline	97.33	86.61
	SCD	**97.55**	87.05
	CCD	97.45	**87.20**
FixResNet50	Baseline	97.16	84.00
	SCD	97.22	83.78
	CCD	**97.39**	**84.85**
ResNeSt50	Baseline	97.74	85.38
	SCD	**97.83**	86.58
	CCD	97.82	**87.12**

Activation Quantify. We quantitatively compare the average magnitude of feature activation with ResNet18 on CIFAR-10 when trained with and without SCD/CCD. The accuracy of Baseline/SCD/CCD is 95.28%/95.90%/95.41% respectively. We randomly sample 512 images in the test set of CIFAR-10 and collect the feature maps of the 2nd/3rd/4th layers followed with average pooling which gets 128/256/512 feature activation respectively. In Fig. 7, we sort the activation within each layer by descending magnitude, and compute the average of 512 samples in the test set. We observe that activation in the CorrDrop models (both with SCD/CCD blocks) significantly outstrip those normally trained models, which thus demonstrates that our SCD/CCD blocks promote the models to consider a wider variety of features when making predictions.

Activation Visualization. We also utilize CAM to visualize the activation units of ResNet20. From Fig. 8 we see that DropBlock randomly drops some regions in the feature maps, thus forcing the model to focus on a wide range of areas which generates a more distributed representation. In contrast, our CorrDrop regularizes the model by masking out those uncorrelated regions in the feature maps which encourages the model to focus on those meaningful regions for classification (e.g. the main object regions). In general, the activation map generated by our methods owns more compact representation and highly activated regions towards the main objects.

4.8. Weakly supervised object localization

Weakly Supervised Object Localization (WSOL) has gained popularity in recent years which aims to localize the object only with the image-level labels. CAM [35] is the pioneer work of WSOL, it generates the score maps to help localize the object. However, CAM has been criticized for its focus on the small discriminative part of the object which motivates most of works to expand the attention regions and localize better. In Section 4.7, we have utilized CAM to compare the activation of model train with and without CorrDrop. As shown in Fig. 8, spatial-wise CorrDrop can help the model focus much on the meaningful regions, which we believe will improve the performance of WSOL. In this section, we finetune the backbone models with the regularization of SCD blocks, then generate the score map with the help of CAM, and finally evaluate the localization performance.

We follow the setting of [47] and two benchmark datasets, i.e., Caltech-UCSD Birds-200-2011 (CUB) [48] and ImageNet [43], are used for evaluation. *GT-known Loc* is adopted as the evaluation metric, which judges the answer correct when the intersection over union (IoU) between the ground truth bounding box and the estimated box for the ground truth class is 50% or more. As suggested in [47], the box accuracy at the optimal threshold τ is reported for threshold independence. We evaluate five WSOL methods with three ImageNet pretrained backbone architectures, i.e., ResNet50, VGG16 and InceptionV3. We use the batch size of 32, a weight decay of 5e-4, an initial learning rate of 1e-3 which decays

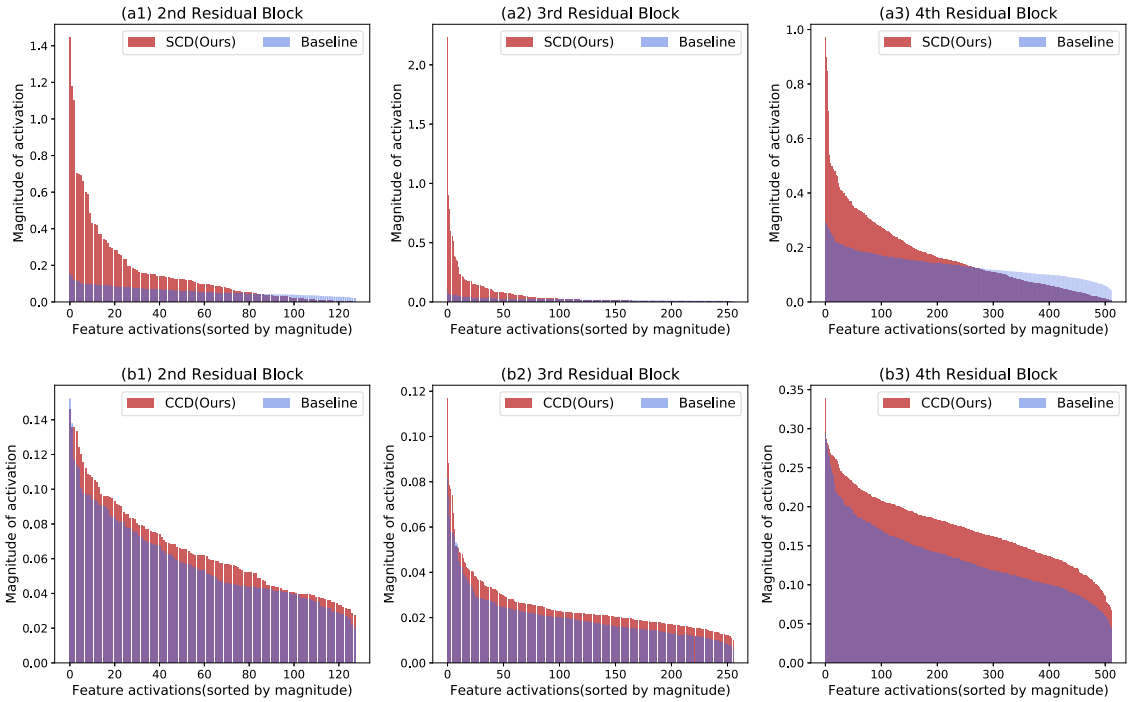

Fig. 7. Magnitude of feature activations in ResNet-18 trained with and without SCD/CCD blocks in different depth of models. The activations are sorted by descending value and averaged with randomly selected 512 test samples in CIFAR-10.

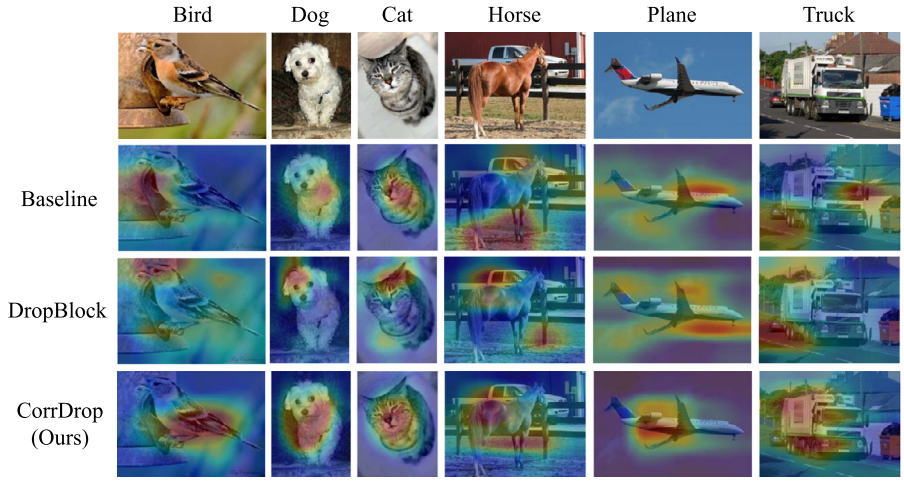

Fig. 8. Class activation mapping (CAM) [35] for ResNet20 trained without regularization, with DropBlock [18] and our CorrDrop.

Table 9
WSOL results on CUB and ImageNet. GT-known Loc metrics(%) are reported. Our proposed CorrDrop(SCD) can consistently improve the performance of localization.

Methods	CUB			ImageNet		
	ResNet50	VGG16	Inception	ResNet50	VGG16	Inception
CAM	66.24	59.72	49.93	60.39	61.42	62.61
ADL	70.12	63.74	54.87	58.87	60.50	61.30
HaS	69.69	73.27	57.63	61.18	61.50	64.23
SCD(Ours)	**71.47**	69.88	60.32	**61.50**	**63.26**	63.61
HaS+SCD(Ours)	69.95	**75.03**	**60.77**	61.13	62.16	**65.13**

by the factor of 10 every 10 and 3 epochs for CUB and ImageNet datasets; the total training epochs is 50 and 10 for CUB and ImageNet, respectively.

Several dropout based WSOL methods tend to drop regions in inputs and intermediate feature maps randomly to diversify the cues for recognition. Here, we compare our method with three state-of-the-arts WSOL methods, i.e., (1) Class activation mapping (CAM) [35], attention-based dropout layer (ADL) [31], and Hide-and-seek (HaS) [49]. Among them, CAM trains a classifier of fully-convolutional backbone with the global average pooling (GAP) structure and weighted the feature maps before the GAP as the score map. ADL introduces a module with a drop mask and importance map which chooses to drop the high-scoring regions or adds attention with importance map. HaS acts as a data augmen-

tation technique and randomly drops the grid patches in the input space. We report the results in Table 9. For both CUB and ImageNet, our method can significantly boost the performance of vanilla CAM and outperforms the compared dropout based WSOL methods. As shown in Table 9, we can gain a larger improvement on the CUB dataset for the reason that CUB is a smaller dataset than ImageNet, and the models trained on CUB are easily to overfit. Furthermore, when SCD is combined with the data augmentation techenique HaS, the performance can be further improved in some cases. These observations demonstrate that our CorrDrop can be easily plugged in the existing CNNs and have the potential applications for other vision tasks.

5. Conclusions

In this paper, we propose a novel and effective structural dropout to regularize CNNs effectively. Unlike the existing regularization methods that may undergo the under-/over-dropping problems for CNNs, our method can remedy such problems by dropping feature based on feature correlation in spatial and channel dimensions. Extensive experiments demonstrate the superiority of our method over other counterparts on different computer vision tasks, network architectures and datasets. Moreover, the visualization of the feature activation map gives us an insight that our method can force the model to learn more compact representations. Beyond the image classification task, we verify the effectiveness of our methods on weakly supervised object localization and reveal the potential applications of our methods in various computer vision tasks. We also show that our methods can be easily plugged into plain CNNs architectures to regularize CNNs. We believe our proposed CorrDrop can work as a general regularization technique in computer vision community.

In future work, we would like to further investigate the effectiveness of our methods on other computer vision tasks such as object detection, semantic segmentation and etc. On the other hand, the visualization of the feature maps in Fig. 8 inspires us to continue exploit the correlation of features to further improve the representational ability of networks.

Declaration of Competing Interest

We declare that we have no financial and personal relationships with other people or organizations that can inappropriately influence our work, there is no professional or other personal interest of any nature or kind in any product, service and/or company that could be construed as influencing the position presented in, or the review of, the manuscript entitled, “Correlation-based Structural Dropout for Convolutional Neural Networks”.

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References

[1] A. Krizhevsky, I. Sutskever, G.E. Hinton, Imagenet classification with deep convolutional neural networks, in: Advances in Neural Information Processing Systems, 2012, pp. 1097–1105.
[2] S. Ren, K. He, R. Girshick, J. Sun, Faster r-cnn: Towards real-time object detection with region proposal networks, in: Advances in Neural Information Processing Systems, 2015, pp. 91–99.
[3] K. He, G. Gkioxari, P. Dollár, R. Girshick, Mask r-cnn, in: Proceedings of the IEEE International Conference on Computer Vision, 2017, pp. 2961–2969.
[4] Y. Kong, M. Feng, X. Li, H. Lu, X. Liu, B. Yin, Spatial context-aware network for salient object detection, Pattern Recognition (2021) 107867.
[5] K. He, X. Zhang, S. Ren, J. Sun, Deep residual learning for image recognition, in: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, 2016, pp. 770–778.
[6] C. Szegedy, V. Vanhoucke, S. Ioffe, J. Shlens, Z. Wojna, Rethinking the inception architecture for computer vision, in: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, 2016, pp. 2818–2826.
[7] G. Huang, Z. Liu, L. Van Der Maaten, K.Q. Weinberger, Densely connected convolutional networks, in: Proceedings of the IEEE conference on computer vision and pattern recognition, 2017, pp. 4700–4708.
[8] Z. Wu, C. Shen, A. Van Den Hengel, Wider or deeper: revisiting the resnet model for visual recognition, Pattern Recognit 90 (2019) 119–133.
[9] A. Krogh, J.A. Hertz, A simple weight decay can improve generalization, in: Advances in Neural Information Processing Systems, 1992, pp. 950–957.
[10] L. Prechelt, Automatic early stopping using cross validation: quantifying the criteria, Neural Networks 11 (4) (1998) 761–767.
[11] Z. Zhong, L. Zheng, G. Kang, S. Li, Y. Yang, Random erasing data augmentation, in: Proceedings of the AAAI Conference on Artificial Intelligence, 2020.
[12] E.D. Cubuk, B. Zoph, D. Mane, V. Vasudevan, Q.V. Le, Autoaugment: Learning augmentation policies from data, in: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, 2019.
[13] S. Ioffe, C. Szegedy, Batch normalization: Accelerating deep network training by reducing internal covariate shift, in: Proceedings of the 32nd International Conference on Machine Learning, 2015.
[14] Y. Li, N. Wang, J. Shi, X. Hou, J. Liu, Adaptive batch normalization for practical domain adaptation, Pattern Recognition 80 (2018) 109–117.
[15] N. Srivastava, G. Hinton, A. Krizhevsky, I. Sutskever, R. Salakhutdinov, Dropout: a simple way to prevent neural networks from overfitting, The journal of machine learning research 15 (1) (2014) 1929–1958.
[16] J. Tompson, R. Goroshin, A. Jain, Y. LeCun, C. Bregler, Efficient object localization using convolutional networks, in: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, 2015, pp. 648–656.
[17] T. DeVries, G.W. Taylor, Improved regularization of convolutional neural networks with cutout, arXiv preprint: 1708.04552 (2017).
[18] G. Ghiasi, T.-Y. Lin, Q.V. Le, Dropblock: A regularization method for convolutional networks, in: Advances in Neural Information Processing Systems, 2018, pp. 10727–10737.
[19] Y. Zeng, T. Dai, S.-T. Xia, Corrdrop: Correlation based dropout for convolutional neural networks, in: ICASSP 2020-2020 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), IEEE, 2020, pp. 3742–3746.
[20] H.S. Baird, Document image defect models, in: Structured Document Image Analysis, Springer, 1992, pp. 546–556.
[21] P.Y. Simard, D. Steinkraus, J.C. Platt, et al., Best practices for convolutional neural networks applied to visual document analysis, Icdar, volume 3, 2003.
[22] F. Cen, X. Zhao, W. Li, G. Wang, Deep feature augmentation for occluded image classification, Pattern Recognition 111 (2021) 107737.
[23] L. Wan, M. Zeiler, S. Zhang, Y. Le Cun, R. Fergus, Regularization of neural networks using dropconnect, in: International Conference on Machine Learning, 2013, pp. 1058–1066.
[24] G. Huang, Y. Sun, Z. Liu, D. Sedra, K.Q. Weinberger, Deep networks with stochastic depth, in: European Conference on Computer Vision, Springer, 2016, pp. 646–661.
[25] B. Zoph, V. Vasudevan, J. Shlens, Q.V. Le, Learning transferable architectures for scalable image recognition, in: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, 2018.
[26] X. Gastaldi, Shake-shake regularization, in: International Conference on Learning Representations, 2017.
[27] R. Keshari, R. Singh, M. Vatsa, Guided dropout, in: Proceedings of the AAAI Conference on Artificial Intelligence, volume 33, 2019, pp. 4065–4072.
[28] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A.N. Gomez, L. Kaiser, I. Polosukhin, Attention is all you need, in: Advances in Neural Information Processing Systems, 2017, pp. 5998–6008.
[29] F. Wang, M. Jiang, C. Qian, S. Yang, C. Li, H. Zhang, X. Wang, X. Tang, Residual attention network for image classification, in: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, 2017, pp. 3156–3164.
[30] S. Zagoruyko, N. Komodakis, Paying more attention to attention: Improving the performance of convolutional neural networks via attention transfer, in: International Conference on Learning Representations, 2017.
[31] J. Choe, H. Shim, Attention-based dropout layer for weakly supervised object localization, in: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, 2019, pp. 2219–2228.
[32] X. Wang, R. Girshick, A. Gupta, K. He, Non-local neural networks, in: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, 2018, pp. 7794–7803.
[33] J. Hu, L. Shen, G. Sun, Squeeze-and-excitation networks, in: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, 2018, pp. 7132–7141.
[34] Y. Yang, Y. Qi, Image super-resolution via channel attention and spatial graph convolutional network, Pattern Recognition 112 (2021) 107798.

[35] B. Zhou, A. Khosla, A. Lapedriza, A. Oliva, A. Torralba, Learning deep features for discriminative localization, in: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, 2016, pp. 2921–2929.
[36] R.R. Selvaraju, M. Cogswell, A. Das, R. Vedantam, D. Parikh, D. Batra, Grad–cam: Visual explanations from deep networks via gradient-based localization, in: Proceedings of the IEEE International Conference on Computer Vision, 2017, pp. 618–626.
[37] A. Prakash, J. Storer, D. Florencio, C. Zhang, Repr: Improved training of convolutional filters, in: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, 2019, pp. 10666–10675.
[38] S. Hou, Z. Wang, Weighted channel dropout for regularization of deep convolutional neural network, in: Proceedings of the AAAI Conference on Artificial Intelligence, volume 33, 2019, pp. 8425–8432.
[39] A. Krizhevsky, G. Hinton, et al., Learning multiple layers of features from tiny images, Technical Report, Citeseer, 2009.
[40] Y. Netzer, T. Wang, A. Coates, A. Bissacco, B. Wu, A.Y. Ng, Reading digits in natural images with unsupervised feature learning (2011).
[41] Y. Le, X. Yang, Tiny imagenet visual recognition challenge, CS231N 7 (2015).
[42] K. Simonyan, A. Zisserman, Very deep convolutional networks for large-scale image recognition, in: International Conference on Learning Representations, 2015.
[43] J. Deng, W. Dong, R. Socher, L.-J. Li, K. Li, L. Fei-Fei, Imagenet: A large-scale hierarchical image database, in: IEEE Conference on Computer Vision and Pattern Recognition, IEEE, 2009, pp. 248–255.
[44] A. Kolesnikov, L. Beyer, X. Zhai, J. Puigcerver, J. Yung, S. Gelly, N. Houlsby, Big transfer (bit): General visual representation learning, in: European Conference on Computer Vision, 2020.
[45] H. Touvron, A. Vedaldi, M. Douze, H. Jégou, Fixing the train-test resolution discrepancy, Advances in Neural Information Processing Systems, 2019.
[46] H. Zhang, C. Wu, Z. Zhang, Y. Zhu, Z. Zhang, H. Lin, Y. Sun, T. He, J. Mueller, R. Manmatha, et al., Resnest: Split-attention networks. arxiv 2020, arXiv preprint: 2004.08955
[47] J. Choe, S.J. Oh, S. Lee, S. Chun, Z. Akata, H. Shim, Evaluating weakly supervised object localization methods right, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2020.
[48] C. Wah, S. Branson, P. Welinder, P. Perona, S. Belongie, The caltech-ucsd birds-200-2011 dataset(2011).
[49] K.K. Singh, Y.J. Lee, Hide-and-seek: Forcing a network to be meticulous for weakly-supervised object and action localization, in: IEEE International Conference on Computer Vision, IEEE, 2017, pp. 3544–3553.

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